UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Wednesday 9 June 2021 13.00 to 16.00 (*3 hours plus any pre-agreed individual adjustment*)

Paper PHA/WP

PHARMACOLOGY WRITTEN PRACTICAL PAPER

*Answer **all four** questions. Each **question** is worth **20 marks in total**.*

Word process your answers in one of the following formats: .doc, .docx, .rtf or .txt (or with permission, hand write)

*Extended Answer questions (2) and (4) have a **750 word** limit*

Provide each main question answer in a single separate document, one for Q1, one for Q2, one for Q3, one for Q4, so four documents in total, named Candidate number_Question number, e.g. 7850X_Q1.docx

Where necessary to perform, show your mathematical workings. You may hand draw mathematical workings (or draw using a tablet), scan or photograph them, and then embed into the corresponding answer document at a suitable point. Clearly label them e.g. "Q1D mathematical working". The same applies to any figures drawn.

You are not required to and should not plot any graphs. You will not be penalised for minor inaccuracies in reading values from graphs provided in questions.

Do not forget to complete the coversheet listing questions answered

Follow the guidance on the University Moodle site on how to upload your documents

Calculator – students are permitted to use an approved calculator

Question 1

A research laboratory has developed a bicyclic peptide technology and has generated a number of ligands against muscarinic acetylcholine receptors. Screening of these ligands in recombinant cell lines identified two candidates suitable for further characterisation in a physiological preparation, termed here Drug 1 and Drug 2. Agonist-induced responses in smooth muscle cells isolated from the rabbit jejunem were detected using a Ca^{2+}-reporter assay and normalised versus the known full agonist, carbachol. Responses were measured following treatment with Drug 1 in the absence or presence of Drug 2. Drug 2 alone had no effect (data not shown). Dose response curves are plotted in **Figure 1**.

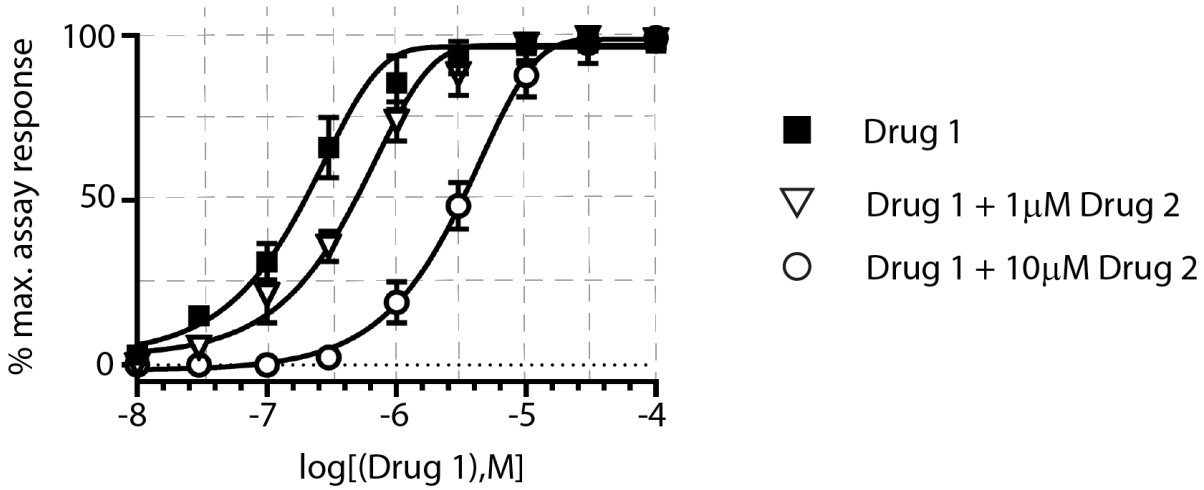

Figure 1. *Agonist-induced dose response curves measured in terms of a Ca^{2+} reporter assay performed on isolated smooth muscle cells from rabbit jejunem.*

A) Comment on the mode of action of Drug 2 (1 mark).

B) Based on Figure 1 can you say if Drug 1 has an intrinsic efficacy of 100 %, i.e. is it bound to and activating 100 % of the receptors? Explain your answer (1 mark)

Using the same assay and cell type three further dose response curves were obtained for Drug 1 at three alternative concentrations of Drug 2 (in addition to the two already obtained and shown in **Figure 1**). The concentrations to achieve half-maximal responses in the presence of all five Drug 2 concentrations were used to establish dose ratios and create **Figure 2**.

[Question 1 continues on the NEXT PAGE]

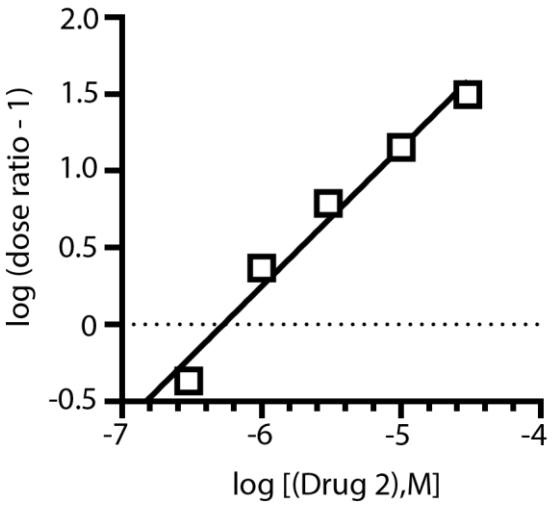

Figure 2. *Relationship of Drug 1 dose ratios to different concentrations of Drug 2.*

C) Calculate the affinity constant of Drug 2. (1 mark)

To further examine the interaction of Drug 1 with the receptor, radioligand binding experiments were performed on the smooth muscle cells isolated from rabbit jejunem. First, specific binding to the cell receptors was measured for the radioligand ^{3}H-NMS (scopolamine methyl chloride) (**Figure 3a**). The maximal specific binding was extrapolated to be 1200 dpm. Next, competition experiments were performed using 5 nM ^{3}H-NMS and increasing concentrations of Drug 1 (**Figure 3b**).

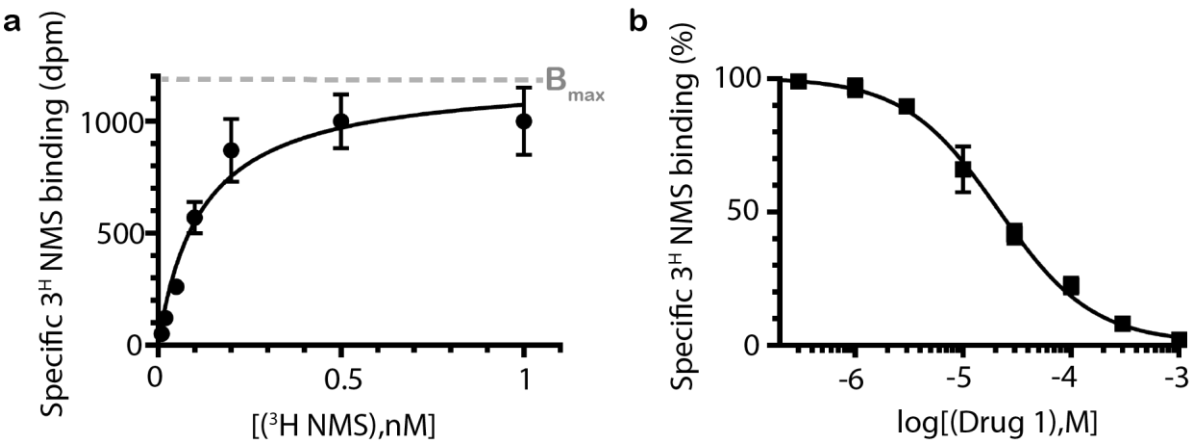

Figure 3. a, *Radioligand binding curve for ^{3}H NMS. Maximum specific binding is extrapolated to be 1200 dpm (grey dash line).* ***b,*** *Competition experiment for 5 nM radioligand ^{3}H-NMS being competed by increasing concentration of Drug 1.*

D) Without drawing any further graphs, calculate the K_A for Drug 1. (4 marks)

[Question 1 continues on the NEXT PAGE]

E) Considering all the data so far, for a situation in which the Drug 1 dose ratio is 5 and Drug 1 achieves a 50 % maximal response, what proportion of receptors are bound by Drug 2? (4 marks)

Next, the investigators tested the bicyclic peptide Drug 1 and Drug 2 in an alternative physiological preparation. The investigators used sinoatrial node cells isolated from rabbit heart in an agonist-driven phosphorylation-reporter assay. As before, responses were normalised against the known full agonist, carbachol. In this alternative cell type either Drug 1 or Drug 2 applied alone exerted an agonistic effect. Dose response curves are plotted in **Figure 4**.

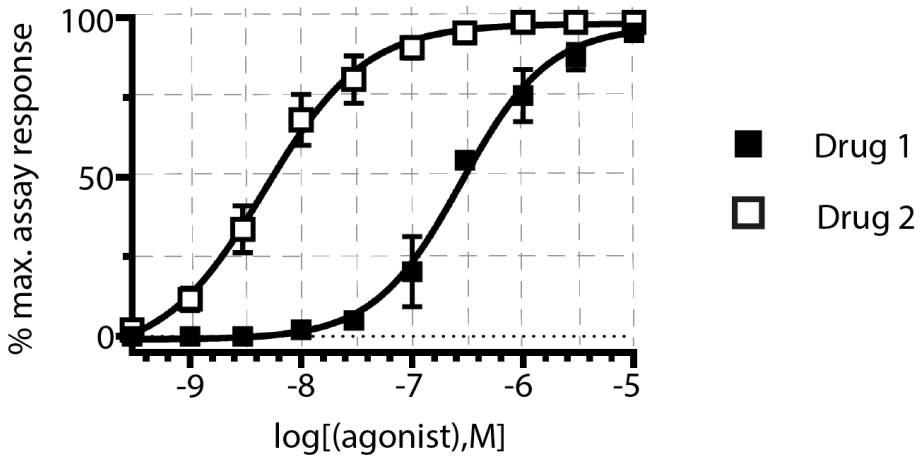

Figure 4. *Dose response curves for drugs measured in terms of a phosphorylation reporter assay performed on isolated sinoatrial cells from rabbit heart.*

The investigators also performed radioligand binding experiments on the isolated cells and determined the dissociation constant of Drug 1 to be 550 nM and of Drug 2 to be 20 nM.

F) Suggest reasons for the difference in action of Drug 2 between the two cell types (2 marks)

G) Perform the appropriate calculations to determine whether Drug 1 or Drug 2 has a higher intrinsic efficacy in the phosphorylation reporter assay. (4 Marks)

H) Which drug would you advise the laboratory to focus on for further therapeutic development, and why? (3 marks)

End of Question 1

2. Explain the factors that determine how effective and selective a drug is at eliciting a specific tissue response. Give examples where possible. (20 marks)

[Question 3 starts on the NEXT PAGE]

Question 3

Sildenafil is used to treat erectile dysfunction. As a free base, sildenafil is a sparingly soluble (0.01 mg/ml water), lipophilic (log P = 2.5) drug that permeates biological membranes rather easily (H-bond acceptors: 10; H-bond donors: 1). The aqueous solubility (4.1 mg/ml) of sildenafil citrate (Viagra®, Pfizer) is much greater than that of the free base, and thus oral dosage forms usually contain sildenafil citrate.

The mean plasma concentration profiles of sildenafil following intravenous and oral administration of 25 mg and 30 mg of Viagra® respectively, to 25 healthy men (40-55 years) are shown in **Figure 1**. The data is also given in Table **1**.

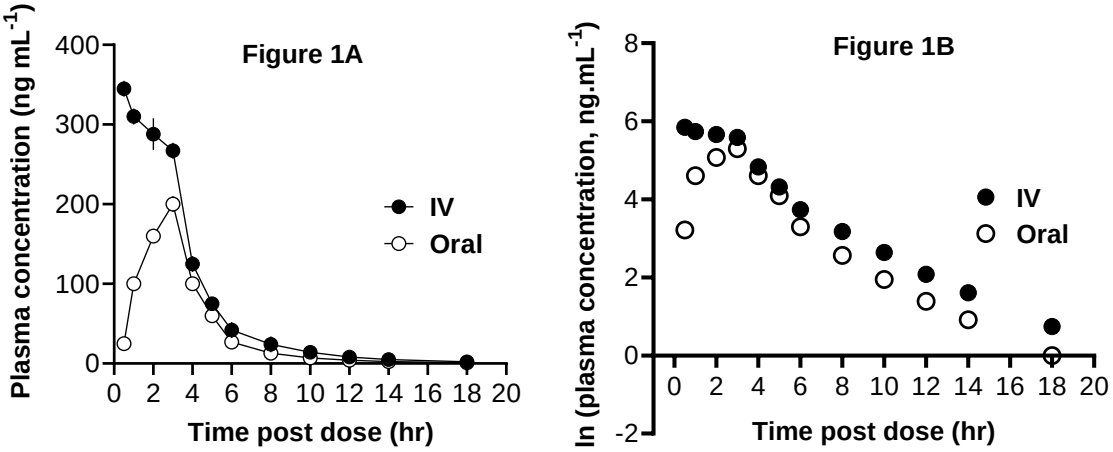

Figure 1. A, *Mean plasma concentration profiles of sildenafil following intravenous (●) and oral (o) administration of 25 mg and 30 mg of Viagra®, respectively to 25 healthy men (40-55 years).* ***B,*** *semi-logarithmic plot of the same data.*

Table 1*. Plasma concentration of sildenafil measured at various time intervals following intravenous (25mg) and oral (30mg) administration of Viagra®.*

Time (hr)	Average plasma concentrations after IV administration		Average plasma concentrations after Oral administration	
	normal conc. ($ng.mL^{-1}$)	semi-log (ln conc.)	normal conc. ($ng.mL^{-1}$)	semi-log (ln conc.)
0.5	345	5.844	25	3.219
1	310	5.737	100	4.605
2	288	5.663	160	5.075
3	267	5.587	200	5.298
4	125	4.828	100	4.605
5	75	4.317	60	4.094
6	42	3.738	27	3.296
8	24	3.178	13	2.565
10	14	2.639	7	1.946
12	8	2.079	4	1.386
14	5	1.609	2.5	0.916
18	2.1	0.742	1	0

[Question 3 continues on the NEXT PAGE]

A) Calculate the absolute bioavailability, half-life and total clearance of orally-administered sildenafil citrate. How can you evaluate the accuracy of the value of the total clearance estimated from the oral administration data? (8 marks).

In studies with radiolabeled ^{14}C sildenafil citrate using isolated rabbit intestine in a perfusion system, 92% of the drug was found to be absorbed by the intestine. Despite this high absorbance, liquid chromatography-mass spectroscopy (LC-MS) data of plasma from whole animals following oral administration showed that much less intact drug was detected in the plasma.

B) What could be the reason for this discrepancy? (1 mark)

In two subsequent studies that took place with two weeks interval in between, the same volunteers were given 75 mg and 100 mg of sildenafil citrate orally and the area under the curves between 0 hr and 24 hrs from the corresponding mean plasma concentration versus time curves were found to be 1534 ng.hr.mL^{-1} and 2045- ng.hr.mL^{-1}, respectively.

C) Based on the information, what can you infer about the kinetics of elimination of sildenafil citrate? Justify your comment. (2 marks)

CB2 pharmaceuticals developed Camerecta® which is a novel formulation of sildenafil citrate suitable for sub-lingual administration and could potentially better suit the elderly patients and/or patients with swallowing problems. A study was conducted with 15 healthy male volunteers (50-60 years old) who took 50 mg of Camerecta® tablet orally (swallowed) in week 1 and the same tablet sub-lingually in week 2. The plasma concentrations of sildenafil and one of its major inactive metabolites, N-desmethyl sildenafil, were measured post-dose at regular intervals for sub-lingual and oral administrations as shown in **Figure 3**.

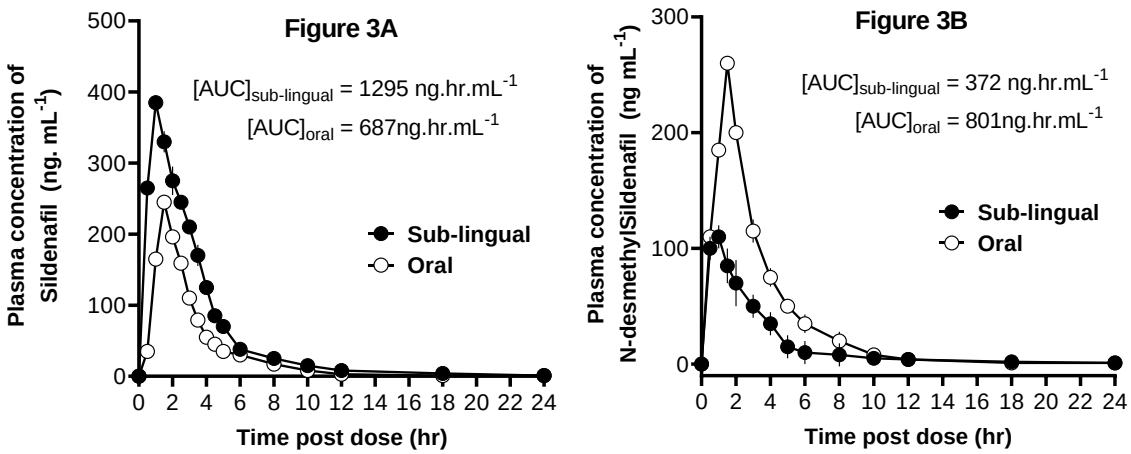

Figure 3. *Plasma concentrations of* ***A,*** *sildenafil and* ***B,*** *one of its major inactive metabolites, N-desmethyl sildenafil, were measured post-dose at regular intervals for sub-lingual (●) and oral (o) administrations.*

D) Determine the bioavailability of sildenafil upon the oral and sub-lingual administration of Camerecta®. Note that the mean plasma concentration versus time plot of sildenafil following intravenous administration of 25mg of Camerecta® to the same volunteers had an area under the curve (0-24hr) of 800ng.h.mL^{-1}. (4 marks)

[Question 3 continues on the NEXT PAGE]

E) Comment on the data. (2 marks)

The following statement was found in the product literature of Viagra®:

"*Do not take this medicine after drinking grapefruit juice. Also consult with your GP or pharmacist before taking this medicine if you have recent history of jaundice or hepatitis and/or taking some other medicines*"

F) Explain why this statement is required. (3 marks)

End of Question 3

4. Describe the factors that influence the oral bioavailability of drugs. (20 marks)

END OF PAPER